EXPERIMENT-02

Aim :-

Convert the ER diagram into a relational schema: Write complete MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Source code :-

```
DROP DATABASE IF EXISTS ecommerce;
CREATE DATABASE ecommerce;
USE ecommerce;
--1. CUSTOMER TABLE
CREATE TABLE Customer (
    Customer_id INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE,
    mob_num VARCHAR(15)
);
--2. ADDRESS TABLE
CREATE TABLE Address (
    Customer_id INT,
    address_type VARCHAR(20),
    PRIMARY KEY (customer_id, address_type),
    FOREIGN KEY (Customer_id)
        REFRENCES Customer(customer_id)
);
--3. SELLER TABLE
CREATE TABLE SELLER (
    Seller_id INT PRIMARY KEY
);
```

```
--4. CATEGORY TABLE
CREATE TABLE Category (
    category_id INT PRIMARY KEY,
    category_name VARCHAR (50) NOT NULL,
    description VARCHAR (200)
);

--5. PRODUCT TABLE
CREATE TABLE Product (
    Product_id INT PRIMARY KEY,
    Product_name VARCHAR (100) NOT NULL,
    description VARCHAR (200),
    price DECIMAL (10,2) NOT NULL,
    seller_id INT,

    FOREIGN KEY (seller_id)
      REFRENCES seller(seller_id)
);

--6. ELECTRONICS TABLE
-- Specialization of product
CREATE TABLE Electronics (
    product_id INT PRIMARY KEY,
    Brand VARCHAR (50),
    Warranty VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
--7. FASHION TABLE
-- Specialization of product
CREATE TABLE Fashion (
    product_id INT PRIMARY KEY,
    size VARCHAR (20),
    material VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product(product_id)
);
```

```
--8. HOME APPLIANCES TABLE
-- Specialization of product
CREATE TABLE Home_Appliances (
    product_id INT PRIMARY KEY,
    power_rating VARCHAR (50),
    capacity VARCHAR (50),
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);
--9. ORDERS TABLE
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    order_date DATE,
    total_amount DECIMAL (10,2),
    discount DECIMAL (10,2),
    customer_id INT,
    FOREIGN KEY (customer_id)
      REFRENCES Customer (customer_id)
);
--10. ORDER_ITEM TABLE
--Weak Entity
-- Composite Primary key
CREATE TABLE Order_Item(
    order_id INT,
    product_id INT,
    quantity INT NOT NULL,
    discount DECIMAL (10,2),
    category_id INT,
    PRIMARY KEY (order_id, product_id)
    FOREIGN KEY (order_id)
      REFRENCES Orders (order_id),
    FOREIGN KEY (product_id)
      REFRENCE product(product_id),
```

```
--11. PAYMENT TABLE
CREATE TABLE Payment (
    payment_id INT PRIMARY KEY,
    Amount DECIMAL (10,2) NOT NULL,
    Customer_id INT,
    seller_id INT,
    FOREIGN KEY (customer_id)
      REFRENCES customer (customer_id),
    FOREIGN KEY (seller_id)
      REFRENCES seller (seller_id)
);
--12. DELIVERY TABLE
CREATE TABLE DELIVERY (
    delivery_id INT PRIMARY KEY,
    tracking_no VARCHAR (50),
    delivery_date DATE,
    estimated_Date DATE,
    product_id INT,
    FOREIGN KEY (product_id)
      REFRENCES Product (product_id)
);

-- INSERT DATA
INSERT INTO customer
 VALUES
 (101, 'Parul', 'parul@gmail.com', '9874540312'),
 (102, 'Aanya', 'aanya@gmail.com', '7943568019'),
 SELECT * FROM customer;

--ADDRESS DATA
INSERT INTO Address
 VALUES
 (101, 'Home'),
```

```
    (101, 'Work'),
    (102, 'Home'),
SELECT*FROM Address;
-- SELLER DATA
INSERT INTO Seller
 VALUES
  (201),
  (202),
SELECT * FROM Seller;
-- CATEGORY DATA
INSERT INTO Category
 VALUES
 (301, 'Electronics', 'Electronics products'),
 (302, 'Fashion', 'Fashion products'),
 (303, 'Home Appliance', 'Home appliance');
-- PRO
 SELECT * FROM Category;
-- PRODUCT DATA
INSERT INTO Product
 VALUES
 (401, 'Laptop', 'Dell Laptop', 55000.00, 201),
 (402, 'T-shirt', 'Cotton T-shirt', 999.00, 202),
 (403, 'Washing Machine', 'Automatic Washing Machine', 30000.00, 201);
 SELECT * FROM Products;
-- ELECTRONICS DATA
INSERT INTO Electronics
 VALUES
  (401, 'Dell', '2 Years');
 SELECT * FROM Electronics;
```

```
-- FASHION DATA
INSERT INTO Fashion
VALUES
(402, 'M', 'cotton');
SELECT * FROM Fashion;
-- HOME APPLIANCES DATA
INSERT INTO Home_Appliances
VALUES
(403, '2000W', '7kg');
SELECT * FROM Home_Appliances;
-- ORDER DATA
INSERT INTO Orders
VALUES
(501, '2026-08-21', 55999.00, 500.00, 101),
(502, '2026-08-21', 30000.00, 1000.00, 102);
SELECT * FROM Orders;
-- ORDER ITEM DATA
INSERT INTO Order_Item
VALUES
(501, 401, 1, 500.00, 301),
(501, 402, 1, 0.00, 302),
(502, 403, 1, 1000.00, 303);
SELECT * FROM Order_Item;
-- PAYMENT DATA
INSERT INTO Payment
VALUES
(601, 55499.00, 101, 201),
(602, 29000.00, 102, 201);
SELECT * FROM Payment;
```

-- DELIVERY DATA

```
INSERT INTO Delivery
VALUES
(701, 'TRK1001', '2026-08-24', NULL, 401),
(702, 'TRK1002', '2026-08-25', NUL, 403);

SELECT * FROM Delivery;
```

OUTPUT:- Customer table

1.

Customer_id	Name	Email	mob_num
101	Parul	parul@gmail.com	9874540312
102	Aanya	aanya@gmail.com	7943568019
NULL	NULL	NULL	NULL

2. Product table

product_id	product_name	description	price	seller_id
401	Laptop	Dell laptop	55000.00	201
402	T-shirt	Cotton T-shirt	999.00	202
403	Washing Machine	Automatic Washing Machine	30000.00	201
NULL	NULL	NULL	NULL	NULL

3. Order_Item table

order_id	product_id	quantity	discount	category_id
501	401	1	500.00	301
501	402	1	0.00	302
502	403	1	1000.00	303
NULL	NULL	NULL	NULL	NULL

4 Delivery table

deliver_id	tracking_no	estimated_date	delivery_date	product_id
701	TRK1001	2026-08-24	NULL	401
702	TRK1002	2026-08-25	NULL	403
NULL	NULL	NULL	NULL	NULL

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